

GOVERNANCE DECLARATION

HHI-GOV-001 | GOVERNANCE CONTINUITY EXPORT BUNDLE | /artifacts

CLASSIFICATION: GOVERNANCE-SENSITIVE

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1.0 PREAMBLE

The Hollow House Institute (hereinafter "the Institute" or "HHI") hereby issues this Governance Declaration as the foundational instrument of its institutional governance architecture. This

Declaration establishes the principles, mandate, scope, and enforcement mechanisms that govern all research, deployment, policy engagement, and publication activities conducted by or under the authority of the Institute.

The Institute recognises that the emergence of IEEE 802.11bf (Wi-Fi Sensing) and adjacent passive radio-frequency inference technologies presents material challenges to civil privacy, individual consent, and the integrity of built environments as spaces free from covert surveillance. These technologies, by their nature, do not require the active participation or awareness of the individuals they observe, and therefore demand governance frameworks of commensurately elevated rigour.

This Declaration is issued in recognition of the gap between the pace of standardisation and deployment of passive sensing technologies and the development of commensurate governance, legal, and ethical frameworks. The Institute resolves to close this gap through independent analysis, public-interest advocacy, and the production of governance artifacts that are freely available, reproducible, and subject to adversarial scrutiny.

This Declaration governs all research activities conducted within the Institute's jurisdiction, all policy positions advanced in the Institute's name, all technology assessments and threat models published under Institute authorship, and all affiliations, partnerships, and vendor engagements entered into by the Institute. No activity of the Institute shall be construed as falling outside the scope of this Declaration unless explicitly exempted by a ratified Amendment issued in accordance with Section 5.0 hereof.

FOUNDATIONAL INSTRUMENT NOTICE

This document constitutes the primary governance instrument of Hollow House Institute. All subsidiary governance documents, operational protocols, research frameworks, and institutional policies derive their authority from and must be consistent with this Declaration.

2.0 INSTITUTIONAL MANDATE

The Hollow House Institute is established as an independent governance research institution. Its mandate is defined as follows:

2.1 Core Functions

The Institute shall perform independent governance research, policy analysis, threat modelling, and public interest advocacy in the domain of radio-frequency (RF)-based sensing and environmental inference technologies. These functions are understood to encompass the full

lifecycle of such technologies, from standardisation and protocol design through to deployment, operational use, data retention, and decommissioning.

2.2 Independence

The Institute operates without commercial affiliation. It accepts no funding, direction, or material influence from commercial entities whose primary interest is the development, sale, or deployment of the sensing technologies within the Institute's scope of analysis. This prohibition extends to indirect relationships, including arrangements with consulting intermediaries, sponsored research agreements, or advisory positions held by Institute personnel in commercial entities operating within the relevant technology domain.

2.3 Open Access Commitment

All findings, governance artifacts, threat models, policy analyses, and research publications produced by the Institute shall be published under open access principles. The Institute shall not impose proprietary restrictions upon its primary research outputs. Governance artifacts, including this Declaration and all documents within the Governance Continuity Export bundle, shall be made available in perpetuity under the Open Governance License identified in the footer of each artifact.

2.4 Scope of Expertise

The Institute's mandate encompasses, but is not limited to: RF sensing standardisation processes (including IEEE 802.11bf and successor standards), passive inference from ambient wireless signals, Channel State Information (CSI) analytics, presence and occupancy detection, behavioural pattern inference, and the intersection of these technologies with data protection law, civil liberties, and built environment policy.

3.0 GOVERNANCE PRINCIPLES

The Institute adopts the following seven (7) core governance principles. These principles are binding upon all Institute activities and shall serve as evaluative criteria for all governance artifacts, research outputs, and policy positions produced or endorsed by the Institute.

3.1 Precautionary Inference Control

Sensing systems operating within the Institute's scope of application shall not infer beyond the stated operational purpose for which they are designed, deployed, or authorised. Any capability of a system to produce inferences beyond its declared operational purpose — whether by design, configuration, model training, or emergent behaviour — constitutes a governance defect

requiring remediation prior to deployment. The Institute shall treat secondary and tertiary inference capability as a primary risk factor in all technology assessments.

3.2 Consent Architecture Primacy

Passive sensing systems, by virtue of their covert operational character, impose upon their operators a heightened consent obligation. The Institute holds that passive sensing systems require affirmative consent frameworks equivalent in rigour and specificity to those applicable to active surveillance technologies. The absence of a perceptible sensing action on the part of a system does not diminish the privacy implications of the inference produced. Consent frameworks must be designed in advance of deployment and must be technically capable of enforcement.

3.3 Provenance Integrity

All governance artifacts produced by or under the authority of the Institute shall carry verifiable chain-of-custody documentation. This documentation shall record the authorship, version history, review record, ratification status, and amendment history of each artifact. Provenance records shall be structured so as to support forensic reconstruction of the governance state of the Institute at any prior point in time. No governance artifact shall be considered authoritative in the absence of a complete and verifiable provenance chain.

3.4 Drift-Safe Structuring

Governance frameworks produced or endorsed by the Institute must include explicit version control and drift detection mechanisms. Governance drift — the unintended divergence of operational practice from documented governance intent — is identified as a primary risk factor in long-duration institutional governance. All frameworks shall specify: a version identifier, a review cadence, criteria for triggering an unscheduled review, and a mechanism by which detected drift triggers a mandatory remediation process.

3.5 Red-Team Accountability

All governance frameworks, policies, and technical standards produced or endorsed by the Institute must undergo adversarial review — commonly referred to as red-teaming — prior to ratification and publication. Adversarial review shall be conducted by reviewers who are independent of the authoring team, and whose mandate is to identify vulnerabilities, inconsistencies, exploitable ambiguities, and foreseeable misuse scenarios within the framework under review. Red-team findings shall be documented, addressed, and retained as part of the artifact's provenance chain.

3.6 Public Interest Primacy

In all cases of conflict between commercial interests and public safety or civil liberties considerations, the Institute shall give primacy to public interest outcomes. Commercial applications of sensing technologies are subordinate to public safety, individual privacy, and civil liberties considerations. The Institute shall not advance, endorse, or fail to critique any governance position that subordinates public interest to commercial convenience, competitive advantage, or commercial deployment timelines.

3.7 Replay Continuity

Governance records maintained by the Institute must support forensic reconstruction of the complete decision history of any governance determination. This principle requires that records be structured, retained, and made accessible in a manner that permits a future reviewer — with no prior institutional knowledge — to reconstruct the full sequence of decisions, deliberations, evidence considered, and amendments applied that resulted in any current governance position. Replay Continuity is a mandatory property of all governance artifacts within the Governance Continuity Export bundle.

4.0 SCOPE OF APPLICATION

This Declaration, and all governance frameworks deriving authority from it, applies to the following technology categories and operational contexts:

Technology / System Category	Scope Notes
IEEE 802.11bf Implementations	All systems implementing or claiming conformance with the IEEE 802.11bf Wi-Fi Sensing standard, including prototype, pre-commercial, and research deployments.
Passive RF Inference Systems	Any system that derives inferences about persons, objects, or environmental conditions from ambient or transmitted radio-frequency

	signals without requiring active participation by the subject of inference.
Wi-Fi CSI Analytics	Systems and analytical pipelines that process Channel State Information (CSI) extracted from 802.11 protocol transmissions for the purpose of environmental or behavioural inference.
Presence Detection Systems	Systems that detect the presence, absence, position, or movement of persons within a defined space using RF-based or passive sensing methodologies.
Behavioural Inference Engines	Systems that classify, predict, or characterise the behaviour, activity, physiological state, or identity of individuals from sensing data, including machine learning models trained on RF or CSI data.
Derivative Sensing Architectures	Any sensing architecture that incorporates, builds upon, or is technically derived from the above categories, operating within HHI research jurisdiction, regardless of the primary stated purpose of the system.

The scope of this Declaration is intentionally construed broadly. Uncertainty as to whether a particular technology or deployment falls within scope shall be resolved in favour of inclusion, consistent with the principle of Precautionary Inference Control set forth at Section 3.1.

SCOPE INTERPRETATION NOTICE

The scope defined in Section 4.0 shall not be narrowed by reference to a technology's stated commercial purpose, target market, or intended user population. A system that is capable of producing inferences within the defined scope is within scope, irrespective of the stated intent of its developers or operators.

5.0 AMENDMENT AUTHORITY

This Declaration may be amended only in accordance with the procedure set forth in this Section. No informal modification, operational deviation, or subsequent policy document shall be construed as amending this Declaration unless it has been ratified in accordance with this procedure.

5.1 Amendment Procedure

Amendments to this Declaration shall follow the four-stage process described below, each stage of which is mandatory and non-waivable:

1. **Written Proposal:** A written amendment proposal must be submitted by a serving Governance Principal of the Institute. The proposal must state: the specific provision(s) to be amended, the proposed amended text, the governance rationale for the amendment, and the anticipated impact on the Institute's existing governance architecture.
2. **Comment Period:** The proposed amendment shall be published and subjected to a mandatory thirty (30) calendar day comment period, during which any affiliated researcher, governance partner, or member of the public may submit written comment. All comments received shall be documented and retained as part of the amendment's provenance record.
3. **Red-Team Adversarial Review:** Prior to ratification, the proposed amendment shall undergo adversarial review in accordance with the Red-Team Accountability principle set forth at Section 3.5. The red-team review record shall be incorporated into the amendment's provenance chain.
4. **Ratification:** The amendment shall be ratified by the signature of a Governance Principal of the Institute. The ratified amendment shall be assigned a document identifier within the HHI-GOV series and incorporated into the Governance Continuity Export bundle.

5.2 First Amendment Reference

The 802.11bf Governance Amendment, designated **HHI-GOV-005**, is recorded as the first amendment exercised under the authority of this Section. HHI-GOV-005 constitutes the initial exercise of the amendment procedure described herein and shall serve as the procedural precedent for all subsequent amendments to this Declaration. Compliance of HHI-GOV-005 with the procedure set forth in Section 5.1 shall be verifiable via the provenance chain of that document.

6.0 COMPLIANCE AND ENFORCEMENT

6.1 Compliance Obligations

The following classes of persons and entities bear binding compliance obligations under this Declaration:

- **Affiliated Researchers:** All individuals conducting research under the Institute's name, with Institute funding, or under Institute supervision, regardless of employment or contractual status.
- **Governance Partners:** All institutions or organisations that have entered into a formal governance partnership, co-publication agreement, or collaborative research arrangement with the Institute.
- **Technology Vendors:** Any commercial entity that has engaged the Institute for governance assessment, certification, endorsement, or advisory services in connection with a technology within the scope defined at Section 4.0.

6.2 Compliance Standards

Compliance with this Declaration requires that all covered persons and entities:

- Adhere to the seven governance principles set forth at Section 3.0 in all relevant activities;
- Maintain complete and accurate provenance records for all governance artifacts produced or modified;
- Submit to adversarial review processes when required by the Institute;
- Report to a Governance Principal any activity that may constitute or risk constituting a breach of this Declaration, within five (5) business days of becoming aware of such activity;
- Cooperate fully with any investigation conducted by the Institute into a suspected breach.

6.3 Non-Compliance Consequences

Breach of the obligations set forth in this Declaration may result in the following consequences, applied individually or in combination, at the determination of a Governance Principal:

Consequence	Description
Suspension of Institutional Affiliation	Immediate suspension of the non-compliant party's status as an affiliated researcher, governance partner, or approved vendor, pending investigation and remediation.

Public Disclosure of Breach	Publication of a factual account of the breach, the non-compliant party, and the Institute's response, as part of the Institute's commitment to transparency and open governance.
Referral to Standards Bodies	Formal referral of the breach, with supporting documentation, to relevant standards bodies, regulatory authorities, or professional organisations with jurisdiction over the non-compliant party or the technology in question.

ENFORCEMENT NOTICE

The consequences described in Section 6.3 are not exhaustive. Nothing in this Section limits the Institute's ability to pursue any other remedy available at law or in equity, or to refer a breach to any authority with jurisdiction to act upon it.

7.0 RATIFICATION

This Governance Declaration is hereby ratified as the foundational governance instrument of Hollow House Institute, effective as of the date of signature below.

This Declaration supersedes all prior informal governance statements, position papers, internal policy memoranda, and operational guidelines of Hollow House Institute that have not been formally ratified under this or a prior equivalent procedure. Upon ratification of this Declaration, no prior informal governance statement shall be relied upon as authoritative with respect to any matter within the scope of this Declaration.

Ratified by:

Amy Pierce Bui

Governance Principal

Hollow House Institute

APPENDIX A — DEFINED TERMS

The following terms, where used in this Declaration or in any governance artifact deriving authority from it, shall have the meanings ascribed to them below.

Passive Sensing

The detection, measurement, recording, or inference of information about persons, objects, or environments through the analysis of signals that are not actively emitted for the purpose of sensing, and without any required interaction with, or awareness on the part of, the subject of observation. Passive sensing includes but is not limited to the use of ambient Wi-Fi transmissions, existing RF infrastructure, and environmental reflection characteristics to derive information about a space or its occupants.

Channel State Information (CSI)

A multi-dimensional measurement derived from the physical layer of IEEE 802.11 (Wi-Fi) communications, representing the properties of a radio channel including amplitude, phase, and time-of-flight characteristics across multiple subcarriers. CSI data encodes how a wireless signal propagates from transmitter to receiver, and is exploitable for environmental sensing, presence detection, motion analysis, and behavioural inference by virtue of its sensitivity to physical changes in the propagation environment.

Behavioural Inference

The process of deriving conclusions about the actions, activities, routines, physiological states, emotional states, identities, or intentions of persons from sensing data, including but not limited to RF signals, CSI measurements, and derived features thereof. Behavioural inference includes both rule-based and machine-learning-based analytical methodologies, and encompasses inferences drawn from aggregated or longitudinal data as well as inferences from single observations.

Governance Principal

An individual formally designated by the Institute as holding authority to propose, review, and ratify governance instruments under the amendment procedure set forth in Section 5.0 of this Declaration. A Governance Principal bears personal accountability for the integrity of governance artifacts ratified under their authority and for the enforcement of this Declaration within their area of institutional responsibility.

Drift-Safe Structure

A governance framework or artifact that incorporates explicit mechanisms for detecting, documenting, and responding to governance drift. A Drift-Safe Structure must include, at minimum: a version identifier, a review cadence, criteria for triggering an unscheduled review, and a defined remediation pathway for detected drift. The absence of any of these components renders a governance artifact non-compliant with the Drift-Safe Structuring principle set forth at Section 3.4.

Replay Continuity

The property of a governance record or artifact collection by which a future reviewer, with no prior institutional knowledge and access only to the documented record, is able to reconstruct the complete and accurate sequence of decisions, deliberations, evidence considered, amendments applied, and governance states that produced any current governance position of the Institute. Replay Continuity is a mandatory property of all artifacts in the Governance Continuity Export bundle.

Provenance Chain

The complete, ordered, and verifiable record of authorship, review, modification, ratification, and amendment history associated with a governance artifact. A Provenance Chain must be sufficient to establish, at any point in time: the identity of the author(s) of each version, the date and nature of each modification, the identity and role of each reviewer, the outcome of any adversarial review, and the identity and authority of the ratifying signatory. A broken or incomplete Provenance Chain renders the affected artifact of uncertain authority.